

■ Exploratory Data Analysis (EDA) Task List

Dataset: Automobile_data.csv

Source:

https://github.com/rushabh-mehta/EDA-on-Automobile-Dataset/blob/master/Automobile_data.csv

1. Data Understanding

- Load the data and display initial rows (head).
- Check data types of each column.
- Generate summary statistics for numerical columns.
- Count unique values for categorical columns.
- Identify constant or near-constant columns.
- Check dataset shape and column names.

2. Data Integrity

- Confirm that column names and types match expected schema.
- Check for missing values and inconsistencies.
- Validate referential integrity across tables if applicable.
- Ensure no duplicate rows unless intentional.
- Check for illogical values (e.g., negative ages, future dates).

3. Data Cleaning

- Handle missing values via imputation or removal.
- Handle duplicate records.
- Convert data types as necessary (e.g., datetime conversion).
- Normalize categorical values (consistent case/spelling).
- Identify and treat outliers (IQR, z-score, boxplot).

4. Univariate Analysis

- Visualize numeric distributions (histogram, boxplot, density).
- Compute summary statistics (mean, median, std, skew).
- Analyze categorical features via bar plots and value counts.
- Check for skewness and consider transformations.

5. Bivariate Analysis

- Numeric vs Numeric: correlation, scatterplots.
- Categorical vs Numeric: boxplots, ANOVA.
- Categorical vs Categorical: cross-tabulations, stacked bars.
- Explore relationships with target variable.

6. Multivariate Analysis

- Create correlation matrix and visualize with heatmap.
- Detect multicollinearity using VIF.
- Perform dimensionality reduction (PCA, t-SNE).
- Conduct feature clustering to identify redundancy.
- Explore interaction effects between multiple variables.

7. Leakage Detection

- Check if any variable leaks target information.
- Ensure time-based features don't use future info.
- Avoid overlap between training and test samples.
- Identify overly predictive features suspiciously correlated with target.

8. Temporal & Spatial Analysis

- For temporal data: check trends, seasonality, missing time periods.
- For spatial data: plot geographic distributions, detect spatial clusters.
- If not applicable, document this explicitly.

9. Reporting & Visualization

- Summarize findings with visualizations and insights.
- Document outliers, missingness, and strong correlations.
- Create EDA notebook report with plots and commentary.
- Propose next steps based on EDA findings.